Spatially Grouped Classification of Glacier Mass-Change States across the Hindu Kush Himalaya: A Retrospective ERA5-Land Baseline

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***Abstract*—The majority of machine-learning (ML) studies of glacier mass balance estimate a continuous value, despite the fact that annual glacier mass change throughout the Hindu Kush Himalaya (HKH) is heterogeneous in space and time. This study examines the effectiveness of four popular classifier families in separating mass gain, mass loss, and near balance on a prepared HKH grid using a planning blueprint that was condensed from three base studies. The modelling table holds 23,698 grid-cell–hydrological-year records (697 cells of 0.5°, April–March years 1991–2024) with 17 predictors: glacier area, year and 15 ERA5-Land variables. Classes adhere to set thresholds of ± 0.1 m w.e., resulting in 64.75% loss, 21.00% gain and 14.25% near balance. Logistic Regression, an RBF Support Vector Machine, Random Forest and Gradient Boosting were compared with five-fold StratifiedGroupKFold validation on 25 K-means location clusters. Near balance remained weak (F1 0.35, precision 0.27), but gradient boosting led (mean fold macro-F1 0.5959; pooled out-of-fold macro-F1 0.6037, balanced accuracy 0.6297, macro ROC-AUC 0.8373). The study is retrospective, uses label thresholds lower than the median reported uncertainty, and tunes and reports using the same folds. A diagnostic also shows implausible temperature and wind anomalies in the two years that contain replaced ERA5-Land months. The results are therefore a documented baseline, not an independent generalization estimate.**

***Index Terms*—glacier mass change, Hindu Kush Himalaya, ERA5-Land, multi-class classification, spatial cross-validation, class imbalance, computational complexity**

I. Introduction

Over years to decades, glaciers integrate atmospheric forcing and store freshwater. One important measure of climate change is their mass balance, which is shown here in meters of water equivalent (mwe). Glacier retreat, variations in seasonal runoff, and the development of glacial lakes are all influenced by a persistent negative balance. While studies in High Mountain Asia demonstrate significant regional and temporal variability, global assessments reveal accelerated glacier loss [1]–[4]. Further changes to glacier extent and water resources are predicted by warming scenarios [5]. Long in-situ records and Nepal-focused synthesis offer more regional context [6], [7].

Observation remains difficult in the HKH. Stake measurements cover relatively few glaciers; geodetic elevation differencing is often multi-year; and physically based energy-balance models require inputs and calibration that are difficult to obtain across steep terrain. Gridded products combining glaciological and geodetic evidence provide annual mass-change targets [8], [9], while ERA5-Land supplies spatially continuous climate predictors [10], [11]. Their combination motivates a classification question: can same-year climate and glacier-area features distinguish grid-cell mass loss, gain and near balance, and how does performance change when complete spatial groups are withheld?

Classification is relevant to screening decisions but removes information about within-class magnitude. Label uncertainty is crucial to interpretation because it also necessitates a boundary for near balance. Using 17 predictors and spatially grouped validation, the main goal is to create and assess a repeatable baseline for classifying annual mass-change state across 697 HKH grid cells. Four classifier families are compared, class-specific errors are described, label uncertainty and data quality are examined, and computational cost is reported in supporting analyses.

The study follows an experimental blueprint derived from three base papers [6], [12], [13]. The blueprint informs the paper structure and planned safeguards, but several stronger elements—including uncertainty-defined labels, nested spatial validation and a locked temporal test—were not executed in the primary spatial experiment. Rather than suggesting that planned work was finished, Section III documents these differences.

The empirical contributions are limited to: (i) documenting the prepared HKH cell-year classification table and its processing caveats; (ii) comparing linear, kernel and tree-ensemble classifiers under shared location-cluster folds; and (iii) quantifying label ambiguity and an apparent ERA5-Land discontinuity associated with replaced months [14]. There is no claim to a novel algorithm, causal explanation, forecast, or glacier-level inference. The rest of the paper describes data and methods, reports results and limitations, audits the blueprint, reviews related work, and ends with future work.

II. Related Work and Research Gap

A. Glacier change and regional evidence

Global and regional syntheses demonstrate significant spatial heterogeneity and widespread glacier loss. Hugonnet et al. document accelerated global elevation loss [1]; studies of High Mountain Asia describe glacier health, rising loss rates and spatially contrasting mass-balance behaviour [2]–[4]. Projections link additional warming to future glacier evolution and water resources [5]. Nepal-focused synthesis and in-situ records provide regional context, including observations from Yala and Rikha Samba [6], [7]. Local evidence is supplemented by field and remote sensing studies from other mountain regions [15], [16]. These studies differ in spatial unit, observation basis and target definition, so they motivate but do not directly benchmark the present grid-cell classification.

B. Machine learning for glacier mass balance

Anilkumar et al. compared linear regression with nonlinear learners for point mass balance in the Alps [13]. Gurung et al. examined radiation and monsoon predictors and employed glacier-grouped validation for Central Himalayan mass-balance modeling [12]. Other studies combine ERA5-Land, geodetic targets and ensemble learning across High Mountain Asia and Central Asia [17], [18]. XGBoost performed best [18]. To investigate drivers in the Tanggula Mountains, Xu et al. integrated OGGM, ensemble learning, and linear models [19]. Two recent studies directly address small or sparse data: Sjursen et al. use XGBoost trained on sparse seasonal and annual measurements to predict mass balance on unmonitored glaciers [20], and van der Meer et al. propose a minimal ML model for very small point-balance datasets [21].

The literature emphasizes the need to test transfer beyond related observations while simultaneously supporting nonlinear climate-based modeling. Although its interpretation is debatable, grouped validation can lessen repeated-location leakage: spatial cross-validation is not always an objective measure of map accuracy, and design-based sampling has been recommended in this regard [22]–[24]. In this case, grouped folds do not support a claim about map accuracy, but rather a relative model comparison and a check against repeated-location leakage.

C. Comparison with the three base papers

Table I compares the base studies with this work. According to the blueprint review, none of the three define a three-class target; instead, they either treat mass balance as a numerical quantity or describe it narratively. This statement supports a description of what the base papers do rather than a claim that classification has never been attempted because the review supporting it was narrow and non-systematic.

D. The research gap

The narrow gap is defined as scope, data, and design rather than as the lack of earlier research. What previous research has established is that nonlinear learners can exploit ERA5-Land predictors for mass balance and that validation design matters. A transparent, repeatable comparison of standard classifier families on the *gridded annual product* over the *whole HKH footprint*, with class-wise results, explicit label-uncertainty diagnostics, and a stated computational cost, is still lacking in documentation. This study contributes such a baseline and the diagnostics that qualify it. It does not assert that the HKH scope, classification, or spatial grouping are novel in and of themselves, nor does it claim novelty from the use of any specific algorithm.

III. Blueprint Audit: Planned Commitment, Executed Treatment, and Status

For each of the three base papers, the blueprint is a completed worksheet. It documents anticipated paper components, their existence in every source study, and their planned modification. This section structure—introduction, literature review, context, methodology, results, discussion, limitations, and conclusion—was influenced by it.

The implemented design incorporates context-first reporting from Khadka et al., glacier-aware validation from Gurung et al., and model diversity from Anilkumar et al. [6], [12], [13]. Spatial grouping addresses repeated-location leakage but does not create buffered geographic independence. The paper also follows the blueprint's restrained novelty language and records data limitations rather than inferring unmeasured high-elevation performance.

IV. Study Area and Data Sources

A. Study area

The prepared target grid spans $27.25°$–$39.25°$ N and $67.25°$–$103.75°$ E and contains 697 cells of $0.5°$, each present in all 34 hydrological years (Fig. 1). The Hindu Kush, Karakoram, the Himalayan arc, portions of the Tibetan Plateau border, Tien Shan, and eastern ranges are all included in this footprint. The smoothed outline of the occupied cells is represented by the shaded envelope in Fig. 1. The HKH polygon was not utilized in the analysis, and it is not an official regional boundary. The study is HKH-wide in the sense of this footprint because, although the original file name includes the word "nepal" for historical reasons, the coordinates clearly extend far beyond Nepal.

B. Datasets

Table III lists the datasets used. The target is the Copernicus Climate Change Service (C3S) global gridded annual glacier mass-change product derived from the World Glacier Monitoring Service (WGMS) Fluctuations of Glaciers database [8], [9]. It distributes annual mass change and its uncertainty on a $0.5°$ global grid for every hydrological year since 1975/76, assigns a glacier to the cell that contains its centroid, and warns that the annual product cannot resolve the different Northern and Southern Hemisphere hydrological-year definitions. I confirmed this relationship on all 23,698 rows to within rounding of the stored three-decimal values. The specific value in m w.e. in the prepared table is equal to the total mass change

TABLE I
RESEARCH-GAP COMPARISON BETWEEN THE THREE MANDATORY BASE PAPERS AND THE PRESENT STUDY.

Previous study	Region	Data	Method	Main limitation or gap	Treatment here
Gurung et al. 2025 [12]	Central Himalaya (HMA regions 14–15)	ERA5-Land, topography, geodetic target 2000–2019	Ridge, Lasso, XGBoost, MLP; glacier-grouped folds	Regression target; neighbouring glaciers can still share climate; radiation signal weak; target period fixed	Classification target; location-cluster folds; 1991–2024 annual panel
Khadka et al. 2023 [6]	Nepal	Literature, ICIMOD inventories	Narrative review with own inventory analysis	No predictive model, no uncertainty propagation; few high-elevation stations	Quantitative baseline; reported-uncertainty diagnostic
Anilkumar et al. 2023 [13]	Central Alps	ERA5-Land monthly, WGMS point data	RF, GBR, SVM, ANN vs linear; random 70/30 split	Point regression; random split with repeated sites; temperate Alps	Grid cells across HKH; same four-family design; grouped folds
This study	HKH, 697 cells	C3S/WGMS gridded target, ERA5-Land, glacier area	LR, SVM-RBF, RF, GB; 5-fold SGKF, 25 clusters	Retrospective; same-fold tuning; fixed thresholds; no nested CV or locked test	Disclosed and diagnosed (Sections X–XI)

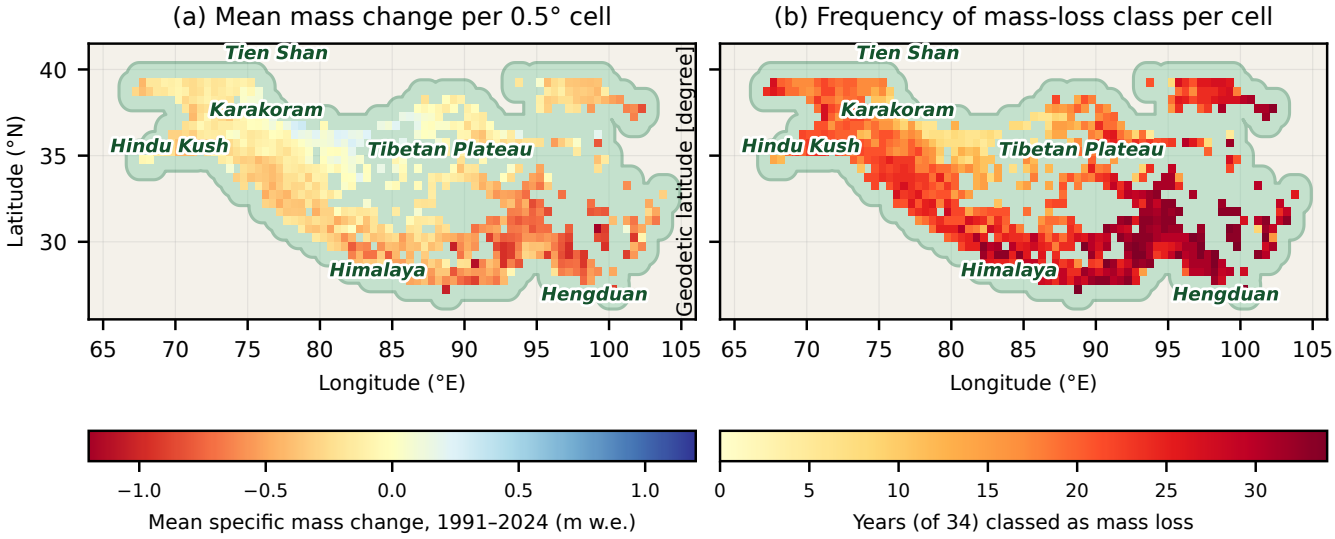


Fig. 1. Study area. (a) Mean specific mass change per 0.5° target cell, 1991–2024. (b) Number of hydrological years (of 34) in which each cell is classed as mass loss. The green-shaded envelope is a smoothed outline of the 697 occupied cells and is not an official HKH boundary. Cells are represented as 0.5° squares over Natural Earth country outlines.

in gigatonnes divided by the water density and the glacier area carried in the same record. The glacier-area column is therefore the glacierised area associated with the cell, not the cell area.

The land component of the ECMWF ERA5 system is replayed with atmospheric forcing to create ERA5-Land, a land-surface reanalysis that includes a lapse-rate correction for the altitude difference between forcing and land grids [10], [11]. Observations influence it only indirectly through that forcing, and the data-set documentation cautions that uncertainty grows backward in time. The inferred 0.1° spacing in the processing notebook agrees with the documented native resolution of about 9 km, which is coarse for steep glacierised terrain and is a limitation discussed later.

Two inputs listed in an earlier plan are not used. Static grids of glacier mask and geopotential and the ICIMOD glacier inventories belong to the Nepal-oriented blueprint, and the final HKH table contains no elevation, aspect or debris-cover information. The only glacier descriptor is glacier area.

V. DATA PROCESSING AND FEATURE ENGINEERING

A. Hydrological-year convention

The target label Y denotes the hydrological year from 1 April of $Y-1$ through 31 March of Y, a half-open interval [1 April $Y-1$, 1 April Y). The C3S documentation gives April to March as the Northern Hemisphere convention [8]. The annual table stores a `time` stamp of 1 January of the label year, which is a label and not the start of the period. There are no incomplete years or imputed years because each of the 23,698 cell-year groups has all 12 distinct months. Month m and calendar year c map to the label

$$Y(c,m) = c + \mathbb{1}[m \geq 4], \tag{1}$$

where $\mathbb{1}[\cdot]$ is the indicator function.

TABLE II
BLUEPRINT AUDIT: PLANNED COMMITMENT, EXECUTED TREATMENT IN THIS PAPER, AND EVIDENCE-BASED STATUS.

Planned commitment	Planned treatment	Executed treatment and evidence	Status
Study scope	Nepal-focused grid and time series	The integration and classification notebooks retain 697 HKH cells and 23,698 cell-years; the executed scope supersedes the earlier Nepal-only plan.	Modified
Target construction	Uncertainty-aware three-class target	Executed target uses fixed ± 0.1 m w.e. thresholds; uncertainty is analysed but does not set labels.	Modified
Classifier diversity	Linear, kernel and ensemble baselines	LR, RBF-SVM, RF and GB are implemented in the spatial notebook; GB extends the initial three-family plan.	Modified
Spatial validation	Grouped folds that keep locations together	25 groups are produced by K-means on 697 coordinates; five folds are assigned to complete groups by StratifiedGroupKFold. Cluster overlap is zero according to assertions.	Implemented
Fold-safe preprocessing	Fit learned transformations on training data	StandardScaler is inside LR/SVM pipelines; GB sample weights are calculated from each training fold.	Implemented
Class imbalance	Reduce majority-class dominance	The final spatial experiment does not use resampling; instead, it uses training-fold sample weights for GB and balanced weights for LR/SVM/RF.	Implemented
Evaluation	Macro-F1, balanced accuracy and class-wise errors	Fold means, pooled OOF metrics, per-class precision/recall/F1, confusion matrices, and one-vs-rest ROC-AUC are all reported in the notebook.	Implemented
Uncertainty sensitivity	Use uncertainty-aware labels or boundary tests	Alternative thresholds and exclusion tests are not used; instead, target uncertainty and near-boundary ambiguity are discussed.	Partial
Temporal holdout	Locked future-year test set	The spatial notebook holds out clusters, not years. A separate chronological notebook has a 2020–2024 test, but it is not the primary deployed experiment.	Partial
Nested evaluation	Tune in inner folds and report outer folds	The same five spatial folds are used in both hyperparameter selection and pooled OOF reporting.	Open
Buffered spatial separation	Exclude nearby training cells around validation blocks	Cluster identity is separated, but no distance buffer or autocorrelation-based block width is implemented.	Open
Feature interpretation	Explain predictor contributions	Correlations and class-wise distributions are reported; the spatial notebook does not perform a supported feature-importance analysis for the primary paper.	Partial
Cost and memory	Repeated timings and memory profiling	Memory profiling and repeated controlled runs are not included; only single-run fit/search timings and model size are recorded.	Partial
Ablations and additional covariates	Test feature groups, topography and other covariates	The topographic additions and listed ablations are not implemented; the primary model keeps 17 predictors.	Open

B. Spatial alignment of ERA5-Land with the target grid

Conservative area-weighted averaging is used to match each 0.5° target cell to the 0.1° ERA5-Land cells that overlap it. The weight of a source cell is proportional to $\Delta\lambda(\sin\varphi_N - \sin\varphi_S)$ for its overlap rectangle, which is the spherical area factor, and the common radius term cancels. At least 70% of a target cell's area must be covered for it to be retained; the minimum coverage was 0.7007, the mean was 0.9888, and the median and upper quartile were 1.0. The lowest-coverage cells lie on the northern edgclose to 39.25° N. The monthly regridded table has 284,376 rows, or 408 months times 697 cells, with no duplicate keys or missing months, and all 697 target cells passed. As a result, the workflow averages source cells that geometrically overlap each target cell rather than interpolating across target cells. Coordinates in the table are the target-cell centres.

C. Predictors and aggregation rules

Glacier area, hydrological year, and 15 ERA5-Land variables make up the 17 predictors (Table IV). Coordinates, the continuous target, its uncertainty, the gigatonne columns and integration metadata are excluded. Dimensionality reduction was not needed to be applied. Aggregation to the hydrological year is variable specific, because accumulated, instantaneous and state variables need different treatment. State variables and winds are day-weighted means of the monthly values. Since the monthly-mean-of-daily-mean ("moda") product expresses accumulated variables as a per-day quantity, radiation, which is expressed as daily accumulated energy, is converted to a mean flux and water-flux variables are summed over months after being multiplied by the number of days in each month. These are the rules that are implemented in the integration notebook and recorded in its metadata table; I treat the rules as implemented but not independently certified, and the metadata audit continues to flag all 15 variables for review against the original NetCDF attributes.

Temperature and melt, snow and snowfall, radiation, runoff, evaporation, and wind are represented by the predictor groups. They motivate the climate feature set but do not make the classifier a surface-energy-balance model. The notebook's aggregation rules are implemented; original NetCDF unit and metadata certification remains open.

TABLE III
DATASETS, ROLES AND SOURCES. RESOLUTIONS ARE THOSE STATED IN THE CITED DOCUMENTATION OR OBSERVED IN THE PREPARED FILES.

Dataset	Provider	Variables used	Spatial res.	Temporal res.	Period used	Role; access
Global gridded annual glacier mass change	C3S; WGMS	Mass change (Gt, m w.e.), glacier area (km^2), uncertainty	0.5°	Annual, hydrological year	1991–2024	Target and glacier area; DOI 10.24381/cds.ba597449 [8]
ERA5-Land	ECMWF / C3S	15 land-surface variables (Table IV)	0.1°(inferred grid step)	Monthly means	Apr 1990 – Mar 2024	Predictors; DOI 10.24381/cds.e2161bac [10], [11]
ERA5-Land recovery product	ECMWF / C3S	12 variables, Sep 2022–Feb 2024	0.1°	Monthly, hour-of-day product	Sep 2022 – Feb 2024	Replacement for affected months [14]
Static glacier area	Inside target product	`glacier_area_km2`	0.5°	Per record	1991–2024	Predictor [8]

TABLE IV
PREDICTORS, PHYSICAL MEANING, SOURCE UNITS AND ANNUAL OPERATION AS IMPLEMENTED IN THE INTEGRATION NOTEBOOK. VALUES IN THE LAST COLUMN ARE TABLE MEANS OVER ALL 23,698 RECORDS.

Name	Group	Meaning	Source unit	Annual operation	Mean
`glacier_area_km2`	Glacier	Associated glacierised area	km^2	Carried from target	114.08
`hydrological_year`	Time	Year label Y	year	Label	–
`t2m`	Ablation	2-m air temperature	K	Day-weighted mean	268.5
`snowc`	Accumulation	Snow cover	fraction; exported scale 0–100	Day-weighted mean	41.7
`sde`	Accumulation	Snow depth	m	Day-weighted mean	1.24
`sd`	Accumulation	Snow water equivalent depth	m w.e.	Day-weighted mean	0.362
`sf`	Accumulation	Snowfall	m w.e. per day	Sum of daily-mean accumulations	0.366
`smlt`	Ablation	Snowmelt	m w.e. per day	Sum of daily-mean accumulations	0.330
`ssr`	Energy	Net surface solar radiation	$J\,m^{-2}$ per day	Mean daily energy as flux	138.2
`str`	Energy	Net surface thermal radiation	$J\,m^{-2}$ per day	Mean daily energy as flux	−79.8
`ro`	Water flux	Total runoff	m per day	Sum of daily-mean accumulations	0.504
`es`	Water flux	Snow evaporation	m w.e. per day	Sum of daily-mean accumulations	−0.035
`sro`	Water flux	Surface runoff	m per day	Sum of daily-mean accumulations	0.268
`e`	Water flux	Total evaporation	m w.e. per day	Sum of daily-mean accumulations	−0.359
`u10`, `v10`	Wind	10-m wind components	$m\,s^{-1}$	Day-weighted mean	0.54, 0.38
`tp`	Accumulation	Total precipitation	m per day	Sum of daily-mean accumulations	0.874

D. The September 2022–February 2024 ERA5-Land issue

ECMWF reported an issue in ERA5-Land monthly averaged reanalysis for September 2022–February 2024, affecting accumulated variables globally [14]. The monthly averaged reanalysis by hour of day at 00:00 is used in the suggested recovery; corrected monthly data are available starting in March 2024. Therefore, before annual aggregation, the impacted accumulations needed to be explicitly handled.

By joining a recovery table on year-month and source coordinates, the integration notebook replaces 12 variables (`t2m`, `sf`, `smlt`, `ssr`, `str`, `ro`, `es`, `sro`, `e`, `u10`, `v10`, `tp`) throughout the 18 impacted months. After the join, its audit reports full replacement coverage and no missing values. Three variables (`snowc`, `sde`, `sd`) remain from the original source. The modeling table keeps all 23,698 records, with seven impacted months falling in HY2023 and eleven in HY2024. No rows were removed.

Scientific equivalency is not established by the join. Since `t2m`, `u10`, and `v10` are not accumulated variables and the accumulation period represented by the 00:00 recovery product was not independently verified, the ECMWF notice does not support substituting hour-specific values for their monthly means. Section IX-G shows anomalies coincident with the replacement. The issue is therefore handled by replacement with unresolved semantics, not certified as corrected.

E. Resulting modelling table

There are no duplicate year-location keys or missing retained predictors in the final table, which has 23,698 rows, 697 locations, and 34 years. Mass change has mean -0.2966 m w.e., median -0.2775 m w.e.and range $[-2.143, 1.181]$ m w.e.; 72.3% of values are negative. Collinearity is strong: r is approximately 1.00 for `sde`/`sd`,

0.98 for ro/tp, 0.97 for ro/sro, and -0.93 for snowc/ssr. These pooled correlations mix spatial and temporal variation.

VI. Problem Formulation

Let $i = 1, \ldots, N$ index records, with $N = 23{,}698$, each defined by a location $\ell_i \in \{1, \ldots, L\}$, $L = 697$, and a hydrological year $Y_i \in \{1991, \ldots, 2024\}$. Let m_i be the specific mass change in m w.e. and u_i its reported uncertainty. The feature vector is $\mathbf{x}_i \in \mathbb{R}^p$ with $p = 17$. The three-class label is

$$y_i = \begin{cases} \text{gain}, & m_i > \tau, \\ \text{loss}, & m_i < -\tau, \\ \text{near balance}, & -\tau \le m_i \le \tau, \end{cases} \qquad \tau = 0.1 \ m\,w.e.. \tag{2}$$

The learning problem is to estimate a classifier $f : \mathbb{R}^p \to \{1, 2, 3\}$ that minimises expected misclassification across classes under a validation scheme that holds out whole spatial groups. With $K = 3$ classes and training set $\mathcal{D}_{\text{tr}}$ of size n, empirical risk minimisation with class weights w_k gives

$$\hat{f} = \arg\min_{f \in \mathcal{F}} \ \frac{1}{n} \sum_{i \in \mathcal{D}_{\text{tr}}} w_{y_i} \, \mathcal{L}(y_i, f(\mathbf{x}_i)) + \Omega(f), \tag{3}$$

where Ω is a regularizer and $\mathcal{L}$ is the model-specific loss.

Since the task is retrospective, the classifier cannot be used before the year ends because $\mathbf{x}_i$ summarizes the same hydrological year as the label. The experiment tests transfer to unseen *places* within the observed period and does not test forecasting because the folds hold out locations rather than years. The product's uncertainty, which has a mean of 0.2635 m w.e. and a median of 0.186 m w.e., is not taken into account by the threshold τ, which is a project convention. Converting a continuous quantity to three bins also discards magnitude, and rows near $\pm\tau$ can flip class under small changes in the target.

VII. Methodology

The executed pipeline is summarised in Fig. 3. It starts from the C3S/WGMS gridded mass-change target and ERA5-Land monthly climate fields, applies the documented preprocessing and spatial/temporal alignment, constructs the 17-feature table, evaluates four classifiers under grouped spatial folds, and saves the selected spatial Gradient Boosting estimator for the Streamlit application. The separate chronological imbalance experiment is not part of the deployed pipeline.

A. Classifiers

The same 17 predictors and five spatial folds were used to fit four classifiers in scikit-learn. A linear baseline, a nonlinear kernel classifier, a bagged tree ensemble, and a boosted tree ensemble were all intended to be tested. The notebook searches 21 configurations in total (3 LR, 2 SVM, 8 RF and 8 GB), giving 105 training-fold fits. Each candidate is ranked by mean validation macro-F1, with balanced accuracy and accuracy used as tie-breakers.

Logistic Regression (LR) is the linear reference. Its pipeline applies `StandardScaler` before multinomial logistic regression, with `class_weight=balanced`, `max_iter=2000` and seed 42. The search varies $C \in \{0.1, 1, 10\}$; the selected setting is $C = 10$. Scaling is fitted separately within each training fold. Although LR is computationally cheap and offers a practical low-capacity comparator, its near-balance F1 (0.20) is the weakest and its pooled OOF macro-F1 is 0.4180.

RBF Support Vector Machine (SVM) provides a nonlinear margin-based comparison. A StandardScaler–SVC pipeline, balanced class weights, $C \in \{1, 5\}$, `gamma=scale`, one-vs-rest decision scores, and `probability=False` are all used. The selected setting is $C = 5$. The notebook uses decision scores for ROC analysis rather than fitting probability calibration. Although it improves on LR, its pooled OOF macro-F1 of 0.4471 is still below both tree ensembles [25].

Random Forest (RF) tests bagged decision trees without feature scaling. It uses `class_weight=balanced_subsample`, parallel workers and seed 42. The training–validation gap and the unrestricted-depth candidates show why depth and leaf-size controls matter in this dataset.

Gradient Boosting (GB) is the selected model. The search varies 100/200 estimators, learning rate 0.05/0.1 and maximum depth 2/3. The winning configuration uses 200 estimators, learning rate 0.1 and depth 3. Since `GradientBoostingClassifier` does not expose a class-weight parameter, the notebook computes balanced per-record sample weights from each training fold and passes them to `fit`. Near-balance F1 is only 0.35, but the pooled OOF macro-F1 is 0.6037, the highest of the four [26]. For application use, the saved model is fitted on all 23,698 records; this final fit is not an independent assessment.

All four models are evaluated on the identical held-out spatial clusters in each fold. The notebook records accuracy, balanced accuracy and macro-F1 during tuning, then creates pooled out-of-fold predictions for class-wise precision, recall, F1, confusion matrices and one-vs-rest ROC-AUC. Because the same folds select and report the models, these values are comparative spatial-CV diagnostics rather than an unbiased independent test estimate.

B. Preprocessing and class-imbalance handling

LR and SVM sit in pipelines with a standard scaler,

$$\tilde{x}_j = \frac{x_j - \mu_j^{\text{tr}}}{\sigma_j^{\text{tr}}}, \tag{4}$$

whose mean μ_j^{tr} and standard deviation σ_j^{tr} are estimated on the training portion of each fold only. Tree models need no scaling.

Cost-sensitive weighting is used to address class imbalance; resampling is not employed. On a training set of size n, the balanced weight of class k is calculated as

$$w_k = \frac{n}{K \, n_k}, \tag{5}$$

where n_k is the class k training record count. RF uses `balanced_subsample`, which recalculates weights within

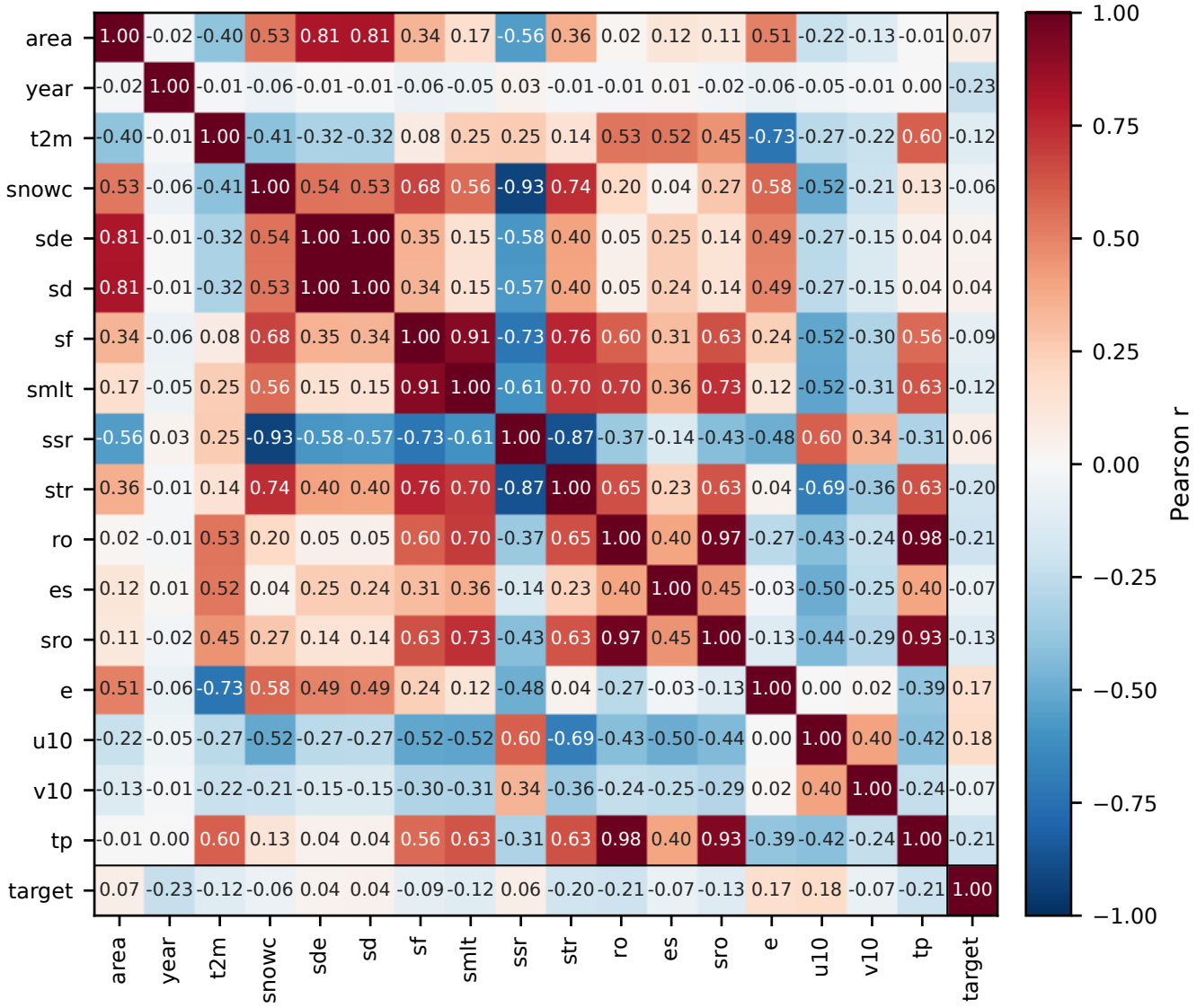


Fig. 2. Pearson correlation among the 17 predictors and the continuous target (last row and column), pooled over all records. The signed Pearson coefficient r (two decimals) is annotated on each cell; a higher $|r|$ denotes a stronger linear association. Collinearity is high within the snow, runoff and radiation groups.

each bootstrap sample, LR and SVM use the library's `balanced` option, and GB gets per-record sample weights calculated from the labels of the current training fold. As a result, weights are only obtained from training data; the validation partition is never resampled or weighted. The final experiment did not include random undersampling, combined schemes, or a no-weighting control, so it is impossible to determine how much weighting helps in comparison to doing nothing.

C. Spatially grouped validation

The 697 distinct coordinate pairs were clustered using k-means [27] with $k = 25$, 20 initializations, and seed 42 after being standardized by axis. The objective is

$$\min_{\{C_c\}} \sum_{c=1}^{25} \sum_{\ell \in C_c} \|\mathbf{z}_\ell - \boldsymbol{\mu}_c\|_2^2 , \tag{6}$$

where $\boldsymbol{\mu}_c$ is the cluster centroid and $\mathbf{z}_\ell$ are the standardized coordinates. One cluster label is inherited by the 34 records at each location. Cluster sizes range from 9 to 40 locations (mean 27.9, median 29). Five-fold StratifiedGroupKFold with shuffling and seed 42 then assigns whole clusters to folds while trying to preserve class proportions [28]. A built-in assertion verifies that no cluster is shared between training and validation in any fold, and it passed for all five folds. The clusters are displayed in Fig. 5. Fold sizes and class counts are listed in Table V.

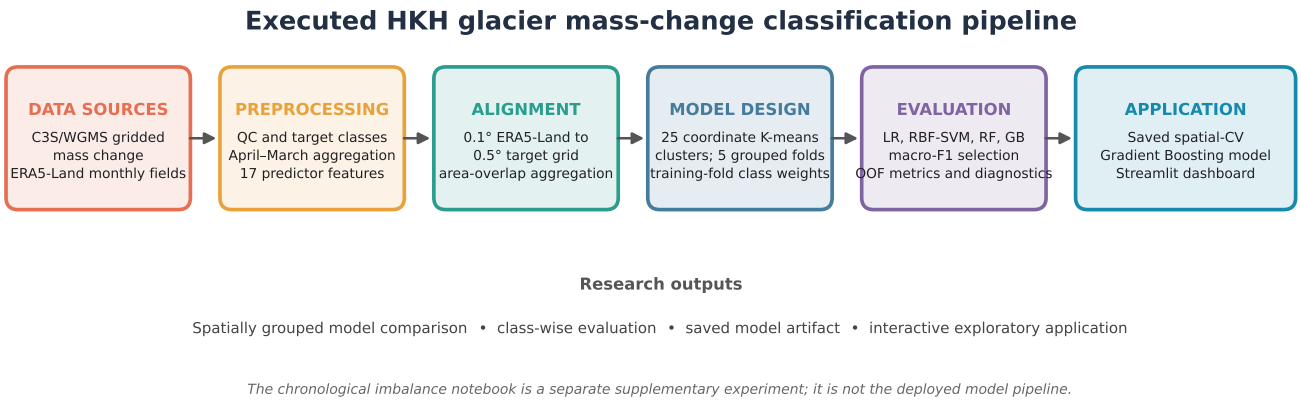


Fig. 3. Executed project workflow from source products to the Streamlit application.

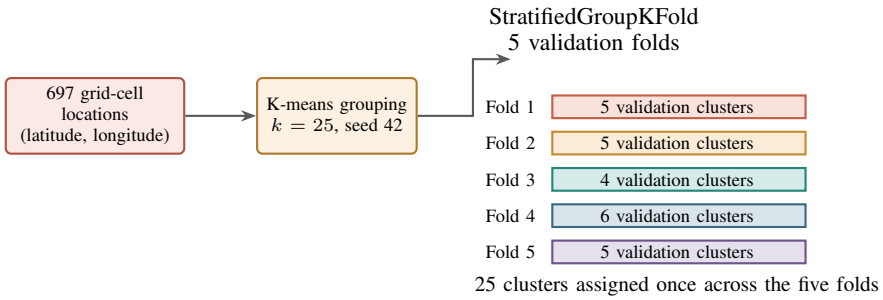


Fig. 4. Spatial cross-validation design reproduced from the executed notebook: the 697 coordinate locations are grouped into 25 K-means clusters, and complete clusters are assigned to five validation folds. The folds contain 5, 5, 4, 6 and 5 validation clusters, respectively. This is cluster-based separation, not a geographic buffer.

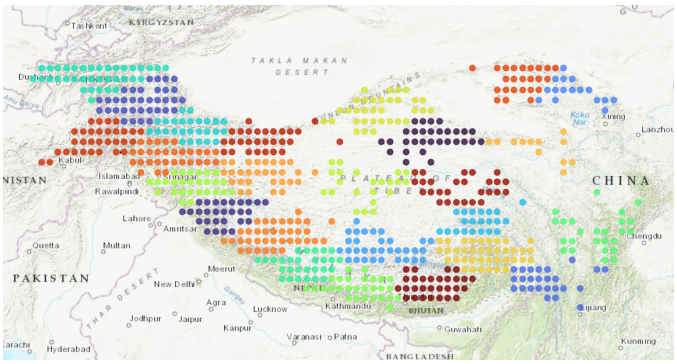


Fig. 5. K-means groups of the 697 grid locations (25 clusters, standardised coordinates). Adjacent clusters may share a color because the color map repeats across clusters. Groups are not buffered geographic blocks; rather, they stop a location or cluster from crossing a fold.

Geographic distance is not separated from cluster identity; adjacent cells may belong to different groups, and standardized latitude/longitude is not projected-distance clustering. A buffer based on target autocorrelation would be stronger; that range was not estimated. Additionally, fold composition varies, with fold 3 having 303 validation gain records and fold 4 having 1,453.

TABLE V
SPATIAL FOLD COMPOSITION (FROM THE VALIDATION NOTEBOOK).

Fold	Train	Val.	Clusters tr/val	Gain	Loss	Near	Shared
1	19,754	3,944	20 / 5	863	2,506	575	0
2	18,904	4,794	20 / 5	1,111	3,022	661	0
3	19,924	3,774	21 / 4	303	3,111	360	0
4	18,122	5,576	19 / 6	1,453	3,192	931	0
5	18,088	5,610	20 / 5	1,246	3,513	851	0

D. Computational complexity

Table VI gives approximate theoretical costs for every algorithm in the final workflow. The expressions disregard implementation constants, parallelism, and caching, are order-of-magnitude, and are based on stated assumptions rather than measured times. In this case, n is the number of training records, p is the number of features, K is the number of classes, I is the number of solver iterations, T is the number of boosting iterations, B trees in a forest, $\tilde{p}$ is the number of candidate features per split ($tildep = \lceil \sqrt{p}, \rceil$ for the classifier default), d is the maximum tree depth, m is the number of support vectors, k clusters, and q is the coordinate dimension.

TABLE VI
THEORETICAL COMPUTATIONAL COMPLEXITY OF THE ALGORITHMS IN THE FINAL METHODOLOGY, WITH SYMBOLS DEFINED IN THE TEXT. ASSUMPTIONS: DENSE DATA, NO PARALLELISM, TREE SPLIT SEARCH BY SORTING, AND PAIRWISE SVM DECOMPOSITION.

Algorithm	Training time	Prediction time (per sample)	Space	Main influencing parameters
Logistic Regression (L-BFGS, multinomial)	$O(I\,n\,p\,K)$	$O(pK)$	$O(np + pK)$	Solver iterations I (cap 2000), C, K, p
SVM, RBF kernel (SMO, pairwise)	Between $O(n^2 p)$ and $O(n^3 p)$	$O(m\,p)$	$O(np)$ data; up to $O(n^2)$ if the kernel matrix is cached; model $O(mp)$	C, γ, class weights; m grows with overlap and noise
Random Forest	$O(B\,\tilde{p}\,n \log n)$ for fully grown trees; $O(B\,\tilde{p}\,n\,d)$ if depth capped	$O(B\,d)$	$O(B\,L_{\max})$ nodes with $L_{\max} \le n/s_{\min}$ for min. leaf size $s_{\min}$	Trees B, depth d, leaf size, features per split
Gradient Boosting (depth-limited trees)	$O(T\,K\,d\,p\,n)$ with presorting-style split search	$O(T\,K\,d)$	$O(T\,K\,2^d)$	Iterations T, depth d, learning rate η (trade-off with T), K
k-means (Lloyd, spatial grouping)	$O(I\,n_{\text{loc}}\,k\,q)$ per initialisation, times 20 initialisations	$O(kq)$ to assign a point	$O(n_{\text{loc}}q + kq)$	Clusters $k = 25$, iterations I, locations $n_{\text{loc}} = 697$, $q = 2$
Standard scaler (LR, SVM)	$O(np)$	$O(p)$	$O(p)$	Features p

These are asymptotic approximations: tree costs depend on split-search implementation and tree shape, whereas SVM costs depend on solver and support vectors [29]. The search cost increases with the number of configurations because 105 fits are needed for the 21 candidates spread across five folds. At the selected settings ($n \approx 19{,}000$, $p = 17$, $K = 3$), SVM has the steepest theoretical training growth; GB uses $T = 200$, $d = 3$, and RF uses $B = 300$. These expressions do not predict measured wall-clock time.

E. Evaluation metrics

In the confusion matrix $\mathbf{M}$, whose entry M_{ab} counts records of true class a predicted as class b, let TP_k, FP_k, and FN_k be the true positives, false positives, and false negatives for class k. The metrics are

$$\text{Acc} = \frac{\sum_k TP_k}{N}, \quad P_k = \frac{TP_k}{TP_k + FP_k}, \tag{7}$$

$$R_k = \frac{TP_k}{TP_k + FN_k}, \quad F1_k = \frac{2P_k R_k}{P_k + R_k}, \tag{8}$$

$$\text{BA} = \frac{1}{K}\sum_{k=1}^{K} R_k, \quad \text{MacroF1} = \frac{1}{K}\sum_{k=1}^{K} F1_k. \tag{9}$$

One-vs-rest ROC-AUC uses each class score against the remaining classes and is macro-averaged. Macro-F1 is the selection metric because it weights classes equally; balanced accuracy and accuracy break ties [30]. Pooled OOF predictions give one held-out-cluster prediction per record.

VIII. EXPERIMENTAL DESIGN

A. Target distribution

Loss accounts for 64.75% of records, so an all-loss classifier reaches 0.6475 accuracy while missing both minority classes. This majority-class baseline is used to interpret accuracy.

TABLE VII
TARGET-CLASS DISTRIBUTION UNDER EQ. (2).

Class	Records	Share (%)
Mass loss ($m < -0.1$)	15,344	64.75
Mass gain ($m > 0.1$)	4,976	21.00
Near balance	3,378	14.25
Total	23,698	100.00

The annual proportions change dramatically: in 2022, there is neither gain nor near-balance cells; in 2023, there are five near-balance cells and no gain; and in 2024, there is one near-balance cell and no gain. The shift is not a temporal holdout but rather a non-stationarity problem because these years occur on both sides of spatial folds.

B. Label uncertainty

The labels and the reported uncertainty are compared in Fig. 8. 66.5% of all records have an uncertainty greater than 0.1 m w.e., and the median uncertainty (0.186 m w.e.) is greater than the class half-width (0.1 m w.e.). In 32.2% of records, including 85.8% of near-balance records, the estimate's absolute value is no greater than its own uncertainty. The distance between the estimate and the closest class boundary is within one reported uncertainty in 41.7% of records, indicating that those records could conceivably belong to a neighboring class. These shares were computed directly from the prepared table. They assume the uncertainty column is a one-sigma interval, which I could not confirm from the supplied documentation.

C. Hyperparameter search and selection

Table VIII lists the compact grids. Using the same five folds for each candidate, the search consists of $3 + 8 + 2 + 8 = 21$

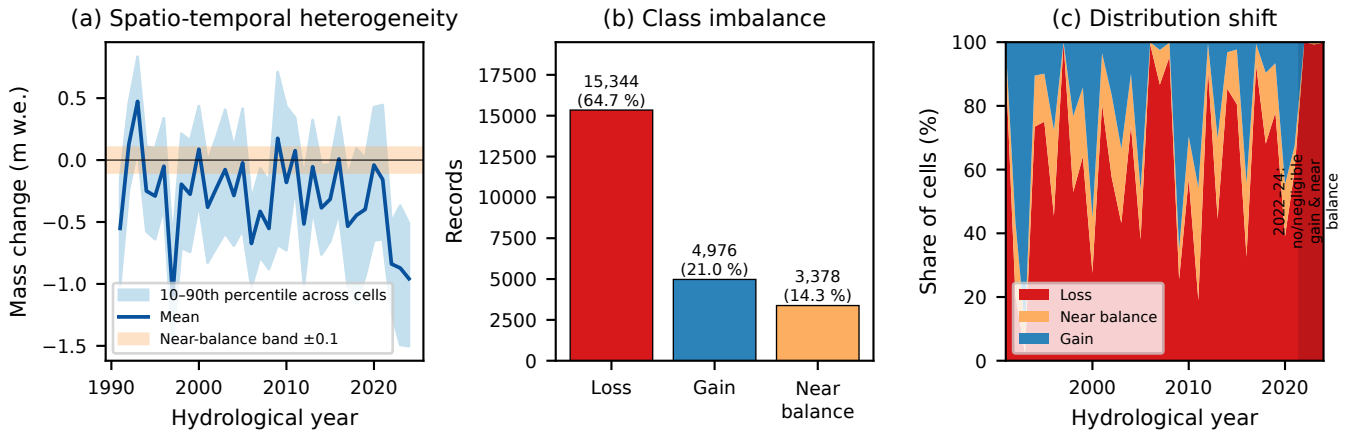


Fig. 6. Problem statement. (a) Annual mass change varies by cell: the near-balance interval is indicated by the shaded strip, and the band covers the 10th to 90th percentile. (b) The three classes are unequally represented. (c) The class mix shifts over time, with gain and near-balance cells almost absent in 2022–2024, so a model trained on all years faces a changing distribution.

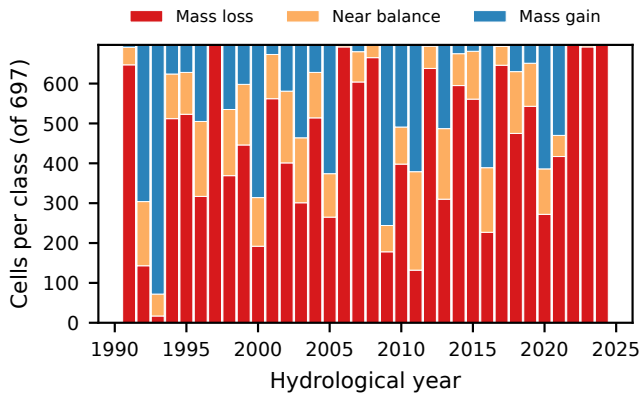


Fig. 7. Cells per class and hydrological year (697 cells per year). The scarcity of gain and near-balance cells after 2021 illustrates temporal distribution shift.

TABLE VIII
SEARCH GRIDS AND SELECTED CONFIGURATIONS (21 CANDIDATES, 105 FITS).

Classifier	Search values (configs)	Selected
LR	$C \in \{0.1, 1, 10\}$ (3)	$C = 10$
SVM (RBF)	$C \in \{1, 5\}$; $\gamma =$ scale (2)	$C = 5$; $\gamma =$ scale
RF	trees 150/300; depth 12/none; leaf 1/3 (8)	300 trees; depth 12; leaf 3
GB	trees 100/200; rate 0.05/0.1; depth 2/3 (8)	200 trees; rate 0.1; depth 3

configurations and 105 fits. The best configuration per family is the one with the highest mean validation macro-F1, with balanced accuracy and accuracy as tie-breakers. The random seed is 42 in every component (clustering, fold assignment, estimators). The analysis used Python with scikit-learn, pandas and NumPy; version numbers are recorded in the experiment metadata and the notebook environment and should be listed in a release.

The grids are small. The same caution applies to search procedures: the compact grid is a practical decision rather than an assertion of optimality because no optimizer outperforms all others over all potential objectives [31]. The optimum may be outside the grid because the chosen SVM cost is the larger of two values and the chosen Gradient Boosting setting is at the upper edge of all three searched ranges (200 trees, rate 0.1, depth 3). LR's low score indicates model capacity rather than under-tuning because it only has one hyperparameter, and the three values yield nearly identical scores (0.4035 to 0.4042 mean fold macro-F1).

D. Reporting protocol

Model selection uses mean validation macro-F1 over the five folds. Pooled OOF predictions are then produced by refitting the selected configuration in each fold. The pooled values are selection-fold diagnostics rather than an independent test since the same folds are used for both reporting and selection. A final Gradient Boosting model was also fitted on all 23,698 records and saved, but its fit is not an evaluation. No test partition, nested loop or chronological holdout exists in this experiment.

IX. RESULTS

A. Model comparison

Gradient Boosting ranked first on every summary (Table IX, Fig. 9). Pooled OOF accuracy was 0.6839 for Gradient Boosting, 0.6476 for Random Forest, 0.5192 for SVM and 0.5119 for Logistic Regression. Balanced accuracy and macro-F1 give the same order (0.6297 and 0.6037 for Gradient Boosting, 0.5382 and 0.5294 for Random Forest, 0.4747 and 0.4471 for SVM, 0.4368 and 0.4180 for Logistic Regression), and so does macro one-vs-rest ROC-AUC (0.8373, 0.7654, 0.6796, 0.6526).

The accuracy of the all-loss rule is already 0.6475. While GB reaches 0.6839, RF (0.6476) is practically tied with it.

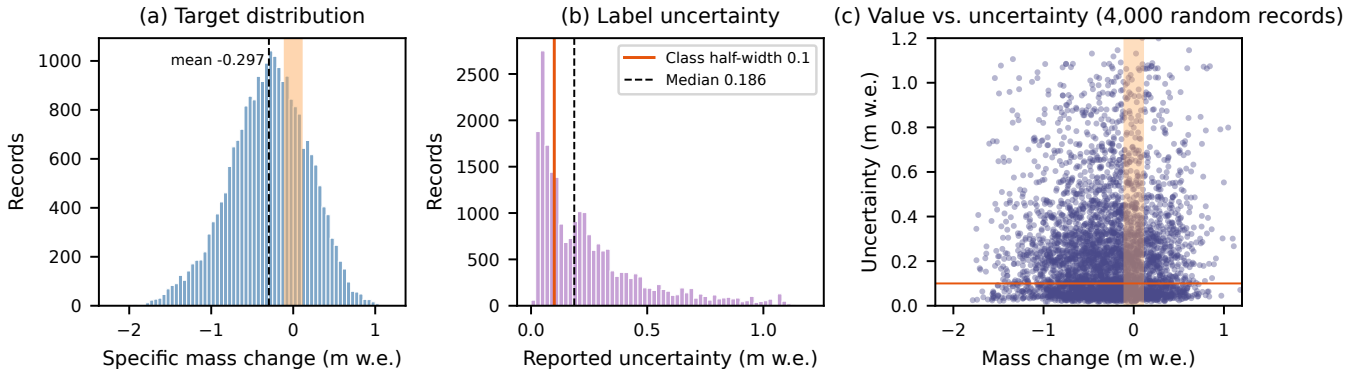


Fig. 8. Target and label uncertainty. (a) Distribution of specific mass change with the near-balance band shaded. (b) Reported uncertainty distribution with the class half-width indicated. (c) Mass change for 4,000 randomly selected records against reported uncertainty; a large number of records fall inside the shaded band with uncertainty above the 0.1 line.

TABLE IX
POOLED SPATIAL OOF METRICS FOR EACH SELECTED CLASSIFIER. SELECTION-FOLD DIAGNOSTICS, NOT INDEPENDENT TEST SCORES.

Classifier	Acc.	Bal. acc.	Macro-F1	Macro AUC	Macro prec.
Gradient Boosting	0.6839	0.6297	0.6037	0.8373	0.6058
Random Forest	0.6476	0.5382	0.5294	0.7654	0.5244
SVM (RBF)	0.5192	0.4747	0.4471	0.6796	0.4621
Logistic Regression	0.5119	0.4368	0.4180	0.6526	0.4251

TABLE X
BEST CONFIGURATION PER FAMILY UNDER FIVE-FOLD SPATIAL-GROUP VALIDATION (MEAN OVER FOLDS; SD OF MACRO-F1 OVER THE FIVE FOLDS).

Classifier	Macro-F1 (SD)	Bal. acc.	Acc.	Train F1
Gradient Boosting	0.5959 (0.0506)	0.6292	0.6881	0.7146
Random Forest	0.5228 (0.0359)	0.5326	0.6557	0.7957
SVM (RBF)	0.4315 (0.0289)	0.4637	0.5259	0.5276
Logistic Regression	0.4042 (0.0263)	0.4304	0.5199	0.4336

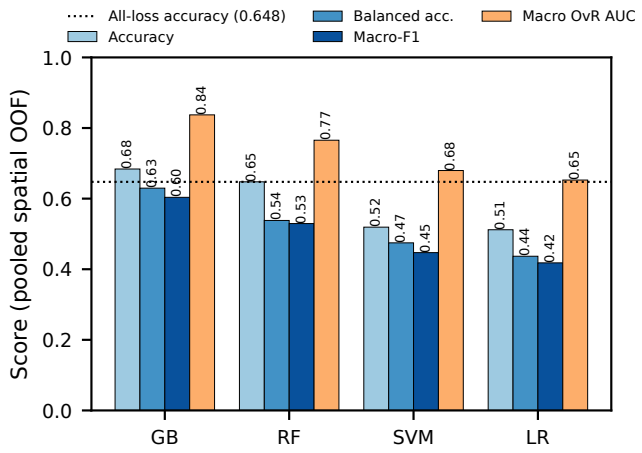


Fig. 9. Pooled spatial OOF accuracy, balanced accuracy, macro-F1 and macro ROC-AUC for the four classifiers. The dotted line marks the all-loss accuracy.

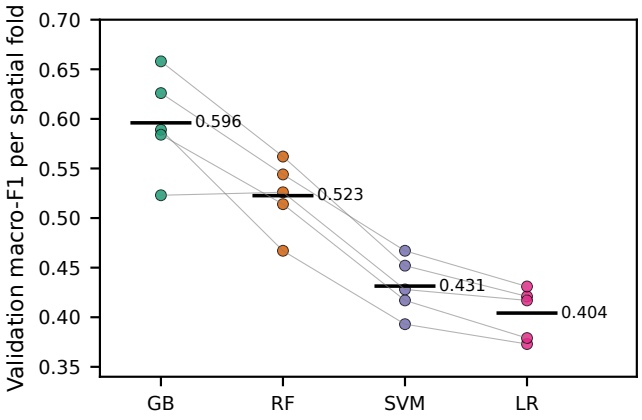


Fig. 10. Validation macro-F1 in each of the five spatial folds for the chosen configurations (dots), with lines joining the same fold across models and fold means (bars). Fold ordering of the models is stable even though the absolute level changes.

Therefore, minority-class performance is obscured by accuracy alone; the more informative ranking is provided by balanced accuracy and macro-F1.

Mean fold macro-F1 was 0.5959 (SD 0.0506) for GB, 0.5228 (0.0359) for RF, 0.4315 (0.0289) for SVM and 0.4042 (0.0263) for LR. Despite significant geographic variation in its fold range (0.523–0.658), GB was ranked first in all five folds.

B. *Class-wise performance*

The headline scores hide the main weakness (Table XI, Figs. 11 and 12). For Gradient Boosting, mass loss has F1 0.81 (precision 0.91, recall 0.73), mass gain 0.65 (precision 0.64, recall 0.66) and near balance 0.35 (precision 0.27, recall 0.50). A precision of 0.27 indicates that roughly three out of four records that were predicted to be near balance actually belong somewhere else, whereas a near balance recall of 0.50 indicates that half of these records were recovered. The

TABLE XI
POOLED SPATIAL OOF CLASS-WISE PRECISION, RECALL AND F1 OF ALL FOUR CLASSIFIERS.

	Mass gain			Mass loss			Near balance		
Classifier	P	R	F1	P	R	F1	P	R	F1
GB	0.64	0.66	0.65	0.91	0.73	0.81	0.27	0.50	0.35
RF	0.51	0.57	0.54	0.82	0.75	0.78	0.24	0.29	0.26
SVM	0.37	0.47	0.42	0.82	0.56	0.67	0.19	0.39	0.26
LR	0.32	0.49	0.39	0.78	0.58	0.67	0.17	0.24	0.20
Support	4,976			15,344			3,378		

TABLE XII
MEASURED WALL-CLOCK COST FROM THE LOGGED SEARCH (INDICATIVE; UNSPECIFIED HOSTED HARDWARE).

Classifier	Fits	Search time (min)	Per fit (min)	Macro-F1 (OOF)
Logistic Regression	15	0.7	0.05	0.4180
Random Forest	40	19.5	0.49	0.5294
SVM (RBF)	10	10.5	1.05	0.4471
Gradient Boosting	40	63.3	1.58	0.6037
Total search	105	94.0	0.90	–

confusion matrix (Fig. 11) shows where the errors come from: of 15,344 true loss records, Gradient Boosting predicted 3,120 (20%) as near balance and 969 (6%) as gain. The majority of the false near-balance predictions, which exceed the 1,687 true near-balance records that were correctly located, are accounted for by those 3,120 records due to the significant loss. Of the 4,976 gain records, 1,352 (27%) were predicted as near balance. According to the label-uncertainty diagnostic, the class is therefore the sink for ambiguity from both sides.

No other model exceeds near-balance F1 of 0.26; recall ranges from 0.24 for LR to 0.50 for GB. Gain precision is also weaker for the linear and kernel models.

C. ROC analysis

One-vs-rest ROC-AUC for Gradient Boosting is 0.900 for gain, 0.886 for loss and 0.726 for near balance (Fig. 13 and Fig. 14). The weakness on near balance is both a ranking weakness and a threshold effect because the three classes' AUC rankings match their F1 rankings. The fact that the gain and loss scores distinguish the classes well over thresholds is a benefit of the model. The two summaries do not have to agree because AUC measures ranking across all thresholds, while precision and recall rely on the classifier's decision rule. In this case, they do.

D. Training versus validation and hyperparameter sensitivity

The macro-F1 gaps between training and validation were 0.27 for RF (0.7957 versus 0.5228), 0.12 for GB, 0.10 for SVM, and 0.03 for LR. Although unrestricted RF trees achieved perfect training fit, their validation scores were significantly lower. GB's searched scores increased with estimator count, learning rate and depth; its optimum lies at the upper edge of the grid, so a broader search is warranted rather than assuming the reported setting is globally optimal.

E. Computational cost

Per-fit times for LR, RF, SVM, and GB are approximately 0.05, 0.49, 1.05, and 1.58 minutes, respectively, according to the single-run notebook log. As a result, GB increased pooled macro-F1 by 0.19 while taking roughly 34 times longer than LR. The final all-record GB fit generated an artifact of 0.75 MB and took 196.67 seconds. These values are merely indicative because hardware was not recorded, memory was not profiled, and timings were not repeated.

F. Predictor behaviour by class

Instead of clear separation, box plots of chosen predictors by class (Fig. 18) display overlap. The 2-meter median annual temperature is 266.6.K represents gain, 267.2 K represents near balance, and 268.0 Although the interquartile ranges heavily overlap, colder cells are more frequently gaining (K for loss). The median snow cover is 37.8 for gain, 38.5 for near balance, and 42.7 for loss. This counterintuitive ordering—more snow cover in loss cells—is only conceivable due to cell differences in elevation and location, and it shows why pooled correlation is not causal. The weak univariate separation is consistent with the modest scores of linear models and with the larger gain from nonlinear interactions.

G. A diagnostic on replaced ERA5-Land months

Fig. 19 plots annual means of three climate variables across the whole grid. Before 2022, annual mean 2-m temperature is 268.54 ± 0.49 K (mean and standard deviation of 1991–2021). In hydrological year 2022, which contains no replaced months, it is 268.75 K ($+0.42$ SD). In 2023, with seven replaced months, it is 267.14 K (-2.85 SD), and in 2024, with eleven replaced months, it is 264.70 K (-7.79 SD). Annual mean zonal wind `u10` falls from 0.54 ± 0.05 to 0.278 and 0.199 $\mathrm{m\,s^{-1}}$, that is -5.5 and -7.1 SD. Annual precipitation, in contrast, stays within one standard deviation of the earlier mean ($+0.60$ and -0.67 SD), and net solar radiation is $+2.15$ SD in 2023 and $+0.72$ SD in 2024.

A 3.8 K annual temperature drop across 697 cells coincides with the replaced months and is not a plausible regional climate signal. The replacement of monthly means for `t2m` and wind with 00:00 values is one possible explanation; however, this hypothesis was not tested against the initial monthly values. HY2023 and HY2024 contain 1,394 records (5.9% of the table) and also have almost no gain or near-balance labels. The classifier could therefore learn an artefactual association between these years, anomalous climate inputs and the loss class. The resulting contamination risk is unquantified.

H. Supplementary chronological imbalance experiment

A second notebook, `glacier_mass_change_integrated_hkh_ml.ipynb`, tests class-imbalance strategies under a different design. It uses 19 predictors, including latitude and longitude, and separates 1991–2019 for development from an untouched 2020–2024 test period. Six expanding-window validation folds use progressively longer training periods. Each of the four algorithms (LR, RBF-SVM,

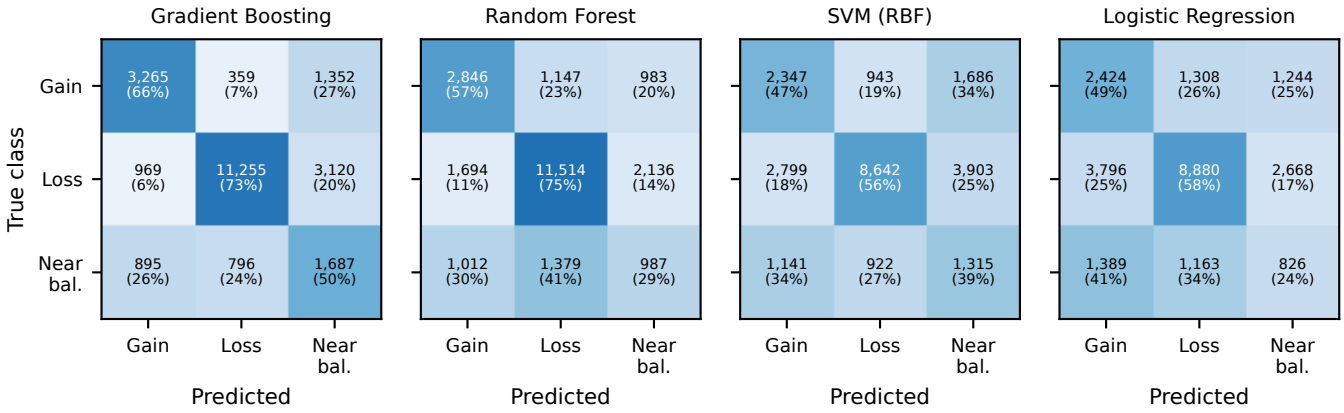


Fig. 11. Pooled spatial OOF confusion matrices. Cells show counts and the row-normalised share of each true class. Counts were read from the validation notebook output and each matrix sums to the class supports in Table XI.

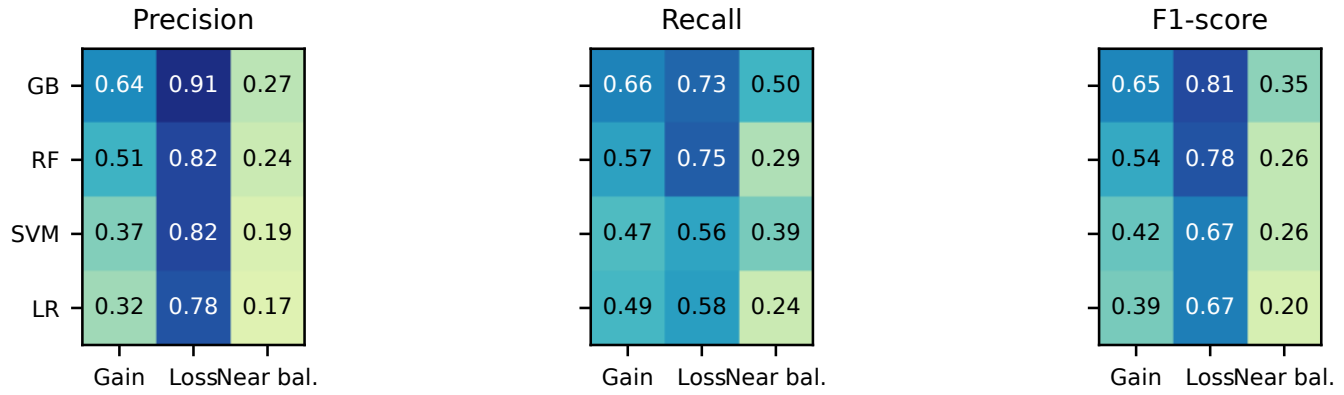


Fig. 12. Class-wise precision, recall and F1 for the four classifiers. Near balance is the weakest class for every model, and precision is the weakest component for that class.

RF, and XGBoost) is tested in four different configurations: neither undersampling nor weighting, undersampling only, weighting only, and both. Four parameter samples per algorithm–strategy combination yield 64 candidates and 384 fold fits. Validation and test records maintain their original class distribution; undersampling is only applied to training folds using `RandomUnderSampler`. Balanced class weights are computed from the original training fold. For XGBoost, those weights are passed as sample weights.

LR with class weighting and no undersampling was chosen because it has the highest mean expanding-window validation macro-F1 (0.3443). Its accuracy is 0.4772, balanced accuracy is 0.3066, and test macro-F1 is 0.2535. The best validation-ranked RF configuration without either imbalance treatment has test macro-F1 0.3587, balanced accuracy 0.3546 and accuracy 0.7145. Instead of changing the chosen model, this reversal is reported: the notebook selects by validation macro-F1 and then assesses the test set. The results also show why the test set must not be used to retrospectively choose a different strategy.

Table XIII summarises the highest validation-ranked configuration within each imbalance strategy. This is a comparison of the best observed strategy–model combinations rather than a controlled causal estimate of the effect of resampling or weighting alone because the algorithm may vary between rows. The notebook uses one fixed seed and only four sampled parameter settings per combination; strategy differences should therefore be treated as exploratory. This experiment is kept separate from the primary spatial experiment: its 19-feature set, temporal split and four-algorithm search are not interchangeable with the spatial notebook's 17-feature, 25-cluster evaluation.

The chosen LR configuration for the chronological experiment is a different outcome and is not used in the provided Streamlit application; the final model for the spatial experiment employs class weighting without undersampling. The application README and model artifact identify the spatial Gradient Boosting estimator as the deployed model.

TABLE XIII
CHRONOLOGICAL EXPERIMENT: BEST VALIDATION-RANKED MODEL CONFIGURATION WITHIN EACH IMBALANCE STRATEGY.

Strategy	Selected algorithm	Val. macro-F1	Test macro-F1	Test balanced acc.	Test accuracy	Practical interpretation
Neither	RF	0.3387	0.3587	0.3546	0.7145	Retains natural training distribution; best test score among the 16 selected configurations, but not the top validation result.
Undersampling only	LR	0.3406	0.2489	0.2976	0.4631	Reduces majority examples in training; may discard useful majority-class variation.
Class weighting only	LR	0.3443	0.2535	0.3066	0.4772	Retains all training observations and penalises minority errors more; highest validation macro-F1 and selected by the notebook.
Both	RF	0.3268	0.3055	0.3344	0.5214	Combines reduced sample size with original-fold weights; can amplify minority influence and performed below the best single treatment here.

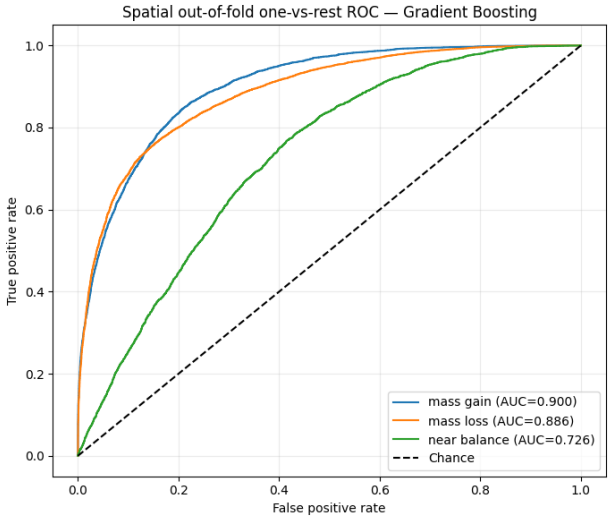


Fig. 13. One-vs-rest ROC curves from pooled spatial OOF scores for Gradient Boosting (notebook output).

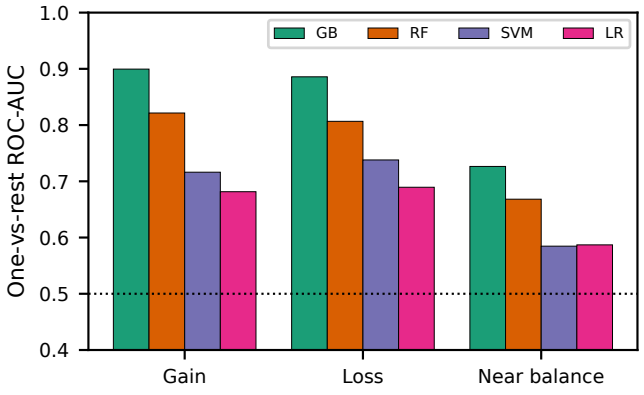


Fig. 14. Class-wise one-vs-rest ROC-AUC for all classifiers. The dotted line marks chance (0.5); near-balance AUC for SVM and Logistic Regression is only 0.58–0.59.

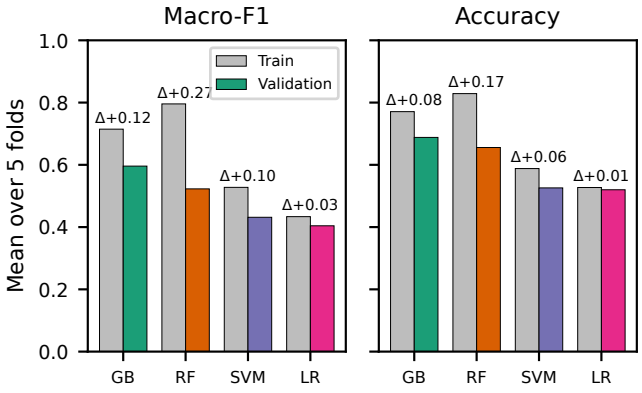


Fig. 15. Mean training and validation macro-F1 and accuracy over five spatial folds for the selected configurations. Labels display the difference between training and validation.

X. DISCUSSION

A. What the evidence supports

Under shared spatially grouped folds, boosted trees give the best separation of the three classes, bagged trees are second, and kernel and linear models are clearly weaker. This ordering agrees in direction with the finding in the Alpine base study that nonlinear and tree-based learners beat a linear baseline and that ensemble trees are strong on medium-sized correlated tabular data [13], and with the use of boosting as the preferred learner in several High Mountain Asia studies [12], [18], [20]. It does not agree on the second place: in the Alpine study the SVM was competitive with the forest, whereas here it was clearly below it. This is a comparison of tendencies rather than numbers due to variations in the task (classification versus point regression), sample size, feature set, and validation. I do not rank this paper against classification scores like macro-F1 and regression scores like R^2 because they are not comparable.

B. Why the weakest model is weak, and why boosting is strongest

Despite having smaller training-validation gaps, LR and SVM have lower scores, which is consistent with their limited ability to interact nonlinearly with correlated climate predictors. Unrestricted depth can cause RF to overfit; depth and leaf-size limits narrow but do not eliminate that gap. GB performs best under the searched shallow-tree settings. Predictor-level attribution is not asserted because no feature-importance analysis was performed for the main spatial experiment. Strong correlations among snow, runoff and radiation variables would

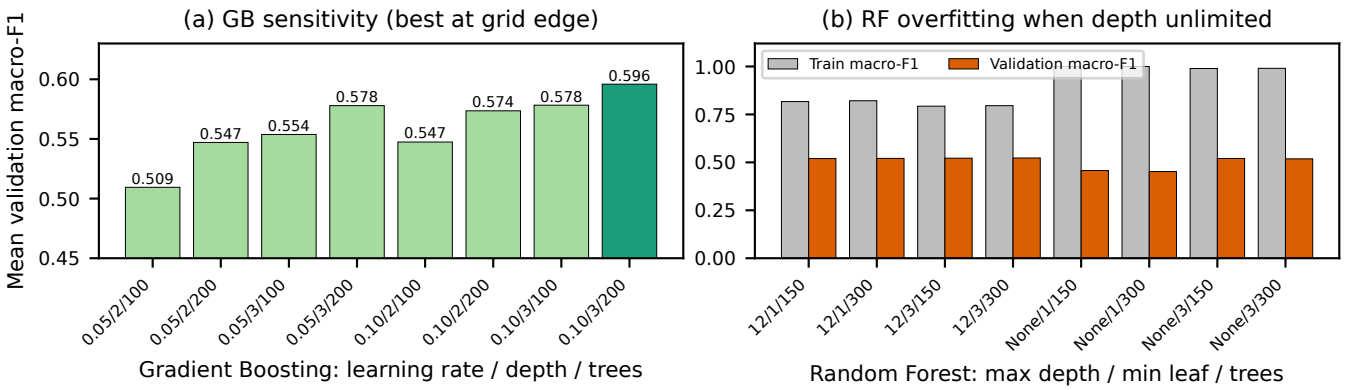


Fig. 16. Hyperparameter sensitivity of mean validation macro-F1. (a) The eight Gradient Boosting configurations; the best lies at the upper edge of the grid. (b) Unlimited depth replicates the training data but not the validation clusters in the eight Random Forest configurations with training and validation scores.

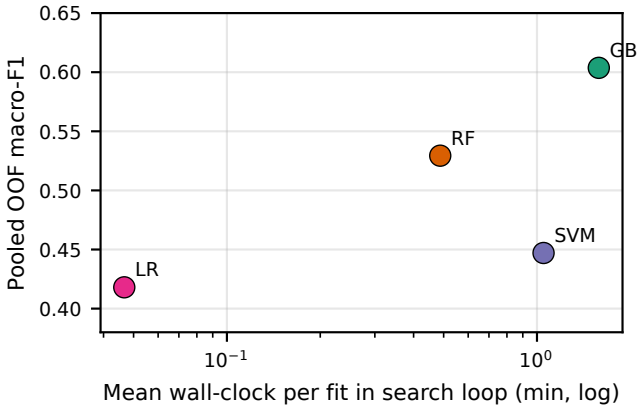


Fig. 17. Performance versus logged cost: mean wall-clock time per fit (log scale) versus pooled OOF macro-F1. Times are derived from a single run on an unidentified piece of hardware.

also make individual importance shares difficult to interpret [13].

C. Near balance is a label problem as much as a model problem

Near-balance weakness may reflect label ambiguity, class size, temporal scarcity and its transitional position between gain and loss. The strongest direct evidence is uncertainty: 85.8% of near-balance records have estimates no larger than their reported uncertainty, whose median is nearly twice the class half-width. Even though Gain is a minority class, imbalance by itself is insufficient because of its higher F1 (0.65). In the past three years, there have only been six near-balance records. These diagnostics suggest an uncertainty-aware label sensitivity analysis, but do not isolate each cause.

D. Spatial validation: what it does and does not show

Group separation is confirmed and repeated-location leakage—the biggest visible leakage in a panel of 697 locations by 34 years—is eliminated by the grouped design. It does not show geographic independence. The clusters are formed in standardised coordinate space, neighbouring clusters touch, and cells near a boundary share climate. Therefore, the apparent difficulty of transfer may be reduced by spatial dependence. Meyer and Pebesma emphasize that cross-validation error only applies inside the model's area of applicability [23], and Wadoux et al. warn that spatial cross-validation cannot replace probability-sample validation for map accuracy [24]. Both points apply here. The folds also include all years, so the test is of spatial transfer within the observed period, and a model that exploits hydrological year can benefit from temporal structure shared by training and validation. A spatio-temporal blocked design, with a buffer and a chronological holdout, would address both limits.

An optimistic bias that increases with the number of configurations compared is introduced by the reuse of folds for reporting and tuning. The effect is probably modest for the small grids here, particularly for LR and SVM, but it is probably larger for Gradient Boosting, where eight configurations are compared and the best of them is reported. This implies that the winning configurations' absolute scores are upper estimates, but it does not change the ranking methodology. The pooled scores are also computed from predictions of models that were selected on the same data.

E. Implications for HKH monitoring

With F1 scores of 0.81 and 0.65, the classifier recovers mass-gain and mass-loss cells in held-out clusters, but it is still unreliable for near balance. This is retrospective cell-level screening rather than forecasting or glacier-level inference because inputs summarize the entire target year and labels represent grid cells.

F. Relation to the blueprint

The primary target is not uncertainty-aware, the primary experiment lacks a locked temporal test, and the executed scope is HKH rather than Nepal-only, as the blueprint revealed. Priority follow-up is to verify and rerun the climate

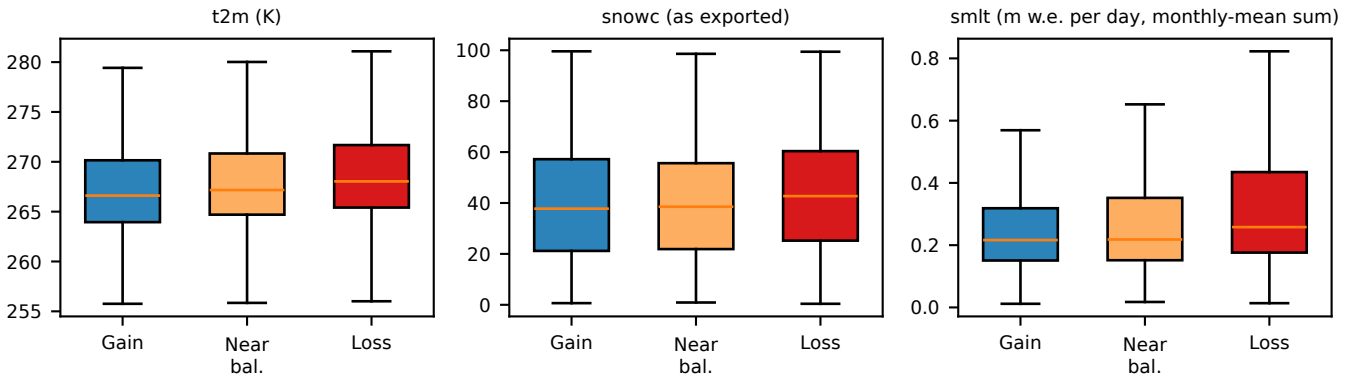


Fig. 18. Distribution of annual 2-m temperature, snow cover and snowmelt by target class (outliers hidden). Distributions overlap strongly, so no single predictor separates the classes.

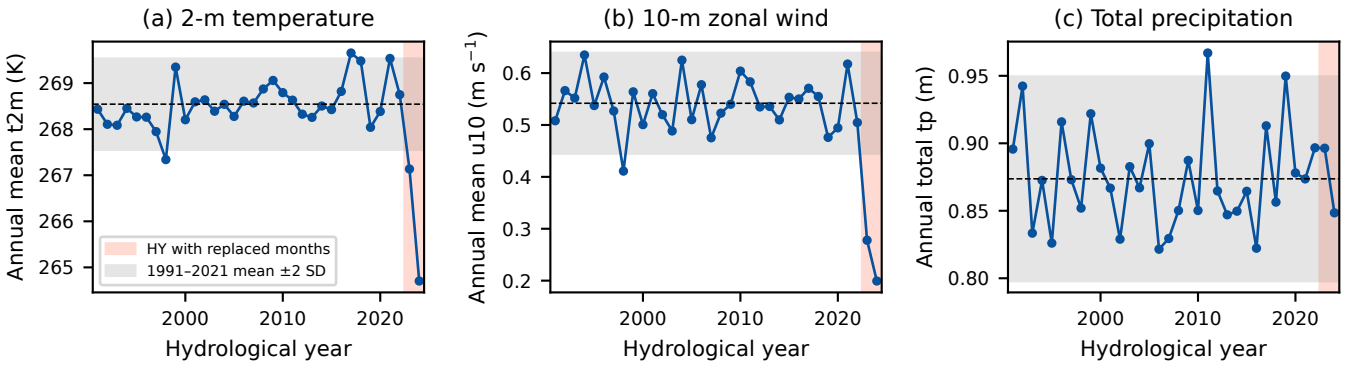


Fig. 19. Grid-mean annual (a) 2-m temperature, (b) 10-m zonal wind and (c) total precipitation by hydrological year. The shaded red band marks the two years that contain replaced months; the grey band is the 1991–2021 mean ±2 standard deviations.

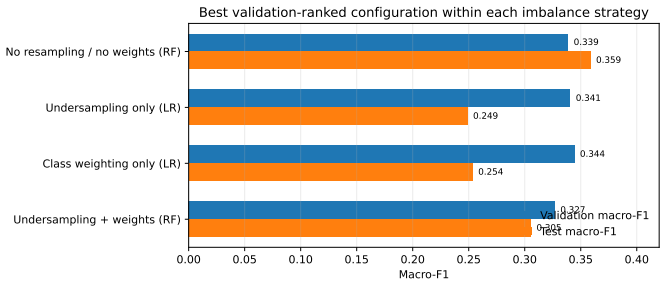


Fig. 20. Validation and 2020–2024 test macro-F1 for the best validation-ranked model within each chronological imbalance strategy. This is not a controlled estimate of the isolated treatment effect; the algorithm varies depending on the strategy.

table, test alternative labels, and implement nested spatial and chronological evaluation.

XI. Threats to Validity and Limitations

Table XIV collects the principal threats by type, with the evidence in this paper and the action that would resolve each.

Claims are limited to the years and cells provided, not to specific glaciers or the HKH as a whole. The uncertainty and climate discontinuity diagnostics are exploratory, and ERA5-Land aggregation rules and the exported snow-cover scale remain unverified against original NetCDF attributes.

XII. Conclusion and Future Work

This study used 23,698 HKH cell-year records to compare LR, RBF-SVM, RF, and GB. In five-fold validation on 25 coordinate clusters, GB led (mean fold macro-F1 0.5959; pooled OOF macro-F1 0.6037; balanced accuracy 0.6297; macro-AUC 0.8373), but near-balance F1 was only 0.35 and its accuracy exceeded the all-loss rule by about 3.6 percentage points. The median reported uncertainty (0.186 m w.e.) exceeds the class half-width, and 85.8% of near-balance estimates are no larger than their uncertainty. In the logged run, GB was also roughly 34 times slower per fit than LR.

These results are retrospective, and the same spatial folds support tuning and reporting. The two years containing replaced ERA5-Land months show implausible anomalies and may contaminate 5.9% of records. Uncertainty-aware labels and nested spatial validation remain open for the primary experiment.

Future work should proceed in this order. (1) Rerun the table and the models, confirm the semantics of the recovery

TABLE XIV
THREATS TO VALIDITY, EVIDENCE IN THIS STUDY AND RESOLVING ACTION.

Type	Threat	Evidence here	Resolving action
Internal	Same folds for tuning and reporting	Selection and pooled OOF scores share folds	Nested spatial CV or untouched spatial region
Internal	Spatial leakage across adjacent clusters	Cluster identity checked, distance not	Buffered blocks; sensitivity to k (e.g., 15, 25, 35)
Internal	Time proxy through hydrological year	Year is a predictor; folds contain all years	Ablate year; chronological holdout
Internal	Grid-edge optima	GB best at upper edge of all three ranges	Extend grid or randomised search
Data	Replaced ERA5-Land months with unverified semantics	-2.85 and -7.79 SD temperature anomalies in 2023, 2024	Keep original `t2m`, `u10`, `v10`; verify units; rerun
Data	Label ambiguity vs. uncertainty	Median $u = 0.186 > 0.1$; 85.8% of near-balance records	Uncertainty-aware labels; boundary exclusion test
Data	Uncertainty definition unknown	1σ vs 1.96σ, annual vs cumulative unconfirmed	Confirm in product documentation
Data	9 km reanalysis over steep terrain; no elevation, aspect or debris	Predictors limited to climate, year and area	Add topographic features; compare with stations
External	Applies to 697 cells and 1991–2024 only	Panel is complete but a single product	Independent product or region
External	Retrospective, not a forecast	Same-year predictors	Lagged or forecast predictors with chronological test
Construct	Class thresholds are a convention	Fixed ± 0.1	Report results for several τ
Computational	Hardware-dependent, single-run timings	Unspecified hosted hardware; no memory profile	Repeated runs on recorded hardware

product for accumulated variables, and restore the original monthly values for non-accumulated variables. (2) Examine labels that are boundary-excluded, uncertainty-aware, and fixed. (3) Add nested spatial validation with a buffer, a cluster-count sensitivity analysis and a chronological holdout. (4) Add topographic predictors and remove the year and area. (5) Add a no-weighting control, expand the Gradient Boosting grid, and assess histogram-based boosting. (6) Repeat the cost measurements on recorded hardware with memory profiling. Until then, the classifier is best read as a documented starting point for screening-level work on HKH glacier mass-change state, not as an operational tool.

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Appendix A
Supplementary Part I: Code and Reproducibility

A. Links

- **Github repository:** https://github.com/Code-With-Samuel/glacier_research
- **Research explanation video (YouTube):** https://youtu.be/brRSi5CpUVU?si=kKkQe97-WWHAX-hf

B. Code snippets

The snippets are taken from the repository notebooks (`supervised_learning_for_classification`, `glacier_mass_change_integrated_hkh_ml`) and the dashboard, shortened only by omitting print statements and figure-generation code.

```python
@st.cache_data(show_spinner="Loading and preparing the HKH dataset…")
def load_data():
    d = pd.read_csv(DATA_PATH)
    d["mass_change_class"] = np.select(
        [d[MASS_COL] > 0.1, d[MASS_COL] < -0.1],
        ["Mass gain", "Mass loss"],
        default="Near balance"
    )
    return d
```

Fig. 21. B.1 Target classes (dashboard loader, `app.py`).

```python
# Standardise coordinate axes before K-means so latitude and longitude
# contribute comparably to the distance calculation.
coord_mean = unique_locations[[LAT_COL, LON_COL]].mean()
coord_std = unique_locations[[LAT_COL, LON_COL]].std(ddof=0).replace(0, 1)
coords_scaled = (unique_locations[[LAT_COL, LON_COL]] - coord_mean) / coord_std

kmeans = KMeans(
    n_clusters=N_SPATIAL_CLUSTERS,
    random_state=RANDOM_STATE,
    n_init=20
)
unique_locations["spatial_cluster"] = kmeans.fit_predict(coords_scaled)
```

Fig. 22. B.2 Spatial clusters (spatial-validation notebook).

```python
cv = StratifiedGroupKFold(
    n_splits=N_SPLITS,
    shuffle=True,
    random_state=RANDOM_STATE
)

folds = list(cv.split(X, y, groups=groups))

fold_summary = []
for fold_no, (train_idx, val_idx) in enumerate(folds, start=1):
    train_groups = set(groups[train_idx])
    val_groups = set(groups[val_idx])
    overlap = train_groups.intersection(val_groups)
    assert not overlap, f"Spatial leakage in fold {fold_no}: {overlap}"

    row = {
        "fold": fold_no,
        "train_rows": len(train_idx),
        "validation_rows": len(val_idx),
        "train_spatial_clusters": len(train_groups),
        "validation_spatial_clusters": len(val_groups),
        "overlapping_clusters": len(overlap)
    }
    for class_id, class_name in enumerate(class_names):
        row[f"validation_{class_name}"] = int(np.sum(y[val_idx] == class_id))
    fold_summary.append(row)
```

Fig. 23. B.3 Grouped folds with leakage check (spatial-validation notebook).

```python
            if model_name == "Gradient Boosting":
                # Calculate balanced weights from THIS training fold only.
                sample_weights = compute_sample_weight(
                    class_weight="balanced",
                    y=y_train_fold
                )
                estimator.fit(
                    X_train_fold,
                    y_train_fold,
                    sample_weight=sample_weights
                )
            else:
                estimator.fit(X_train_fold, y_train_fold)
```

Fig. 24. B.4 Fold-wise balanced weights for Gradient Boosting (spatial-validation notebook).

```python
def fit_one_model(algorithm, params, X_train, y_train,
                  undersampling=False, class_weighting=False, seed=RANDOM_STATE):
    y_train_array = np.asarray(y_train, dtype=int)
    weights_by_class = (
        original_fold_class_weights(y_train_array) if class_weighting else None
    )

    if undersampling:
        sampler = RandomUnderSampler(random_state=seed)
        X_fit, y_fit = sampler.fit_resample(X_train, y_train_array)
    else:
        X_fit, y_fit = X_train, y_train_array

    estimator = build_estimator(algorithm, params, class_weights=weights_by_class)
```

Fig. 25. B.5 Imbalance settings (chronological notebook).

```python
st.markdown("**2 · Classify the supplied conditions**")
if st.button("Run mass-change classification",type="primary",use_container_width=True):
    x=pd.DataFrame([[vals[f] for f in FEATURES]],columns=FEATURES)
    pred=int(model.predict(x)[0])
    probs=model.predict_proba(x)[0]
    pred_name=CLASS_NAMES.get(pred,str(pred))
```

Fig. 26. B.6 Prediction in the dashboard (`app.py`).

Appendix B
Supplementary Part II: Streamlit Application Demonstration

The dashboard (`app.py`) was running locally for these captures on the supplied data and model. The interactive map pages use an online basemap through Folium; in the capture environment the map component did not render, so no map screenshot is shown.

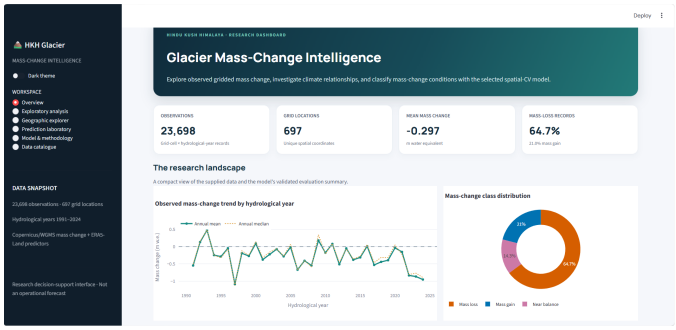


Fig. 27. Streamlit dashboard overview.

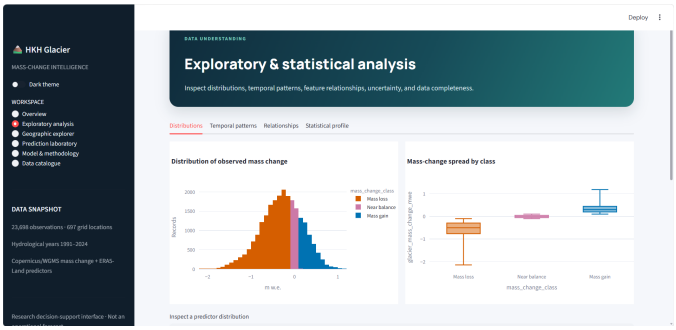


Fig. 28. Streamlit exploratory analysis: class distribution and summary view.

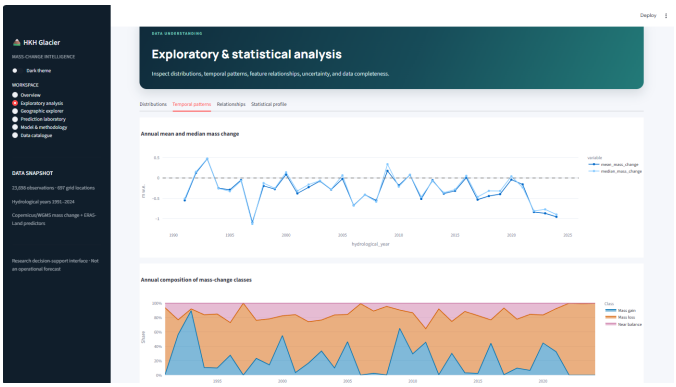


Fig. 29. Streamlit exploratory analysis: predictor comparison view.

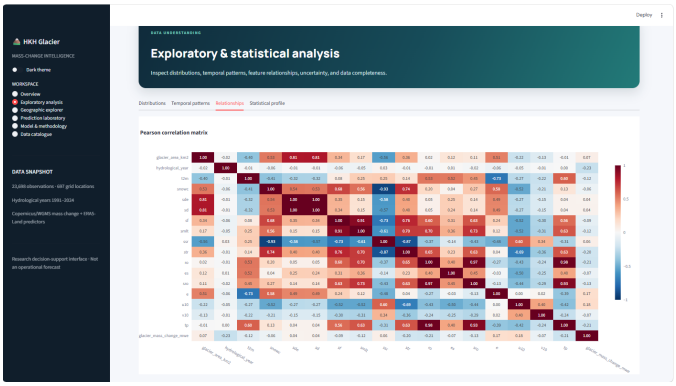


Fig. 30. Streamlit exploratory analysis: additional diagnostic view.

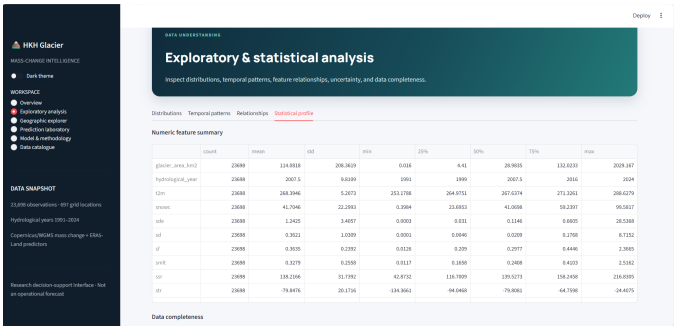


Fig. 31. Streamlit exploratory analysis: distributional diagnostic view.

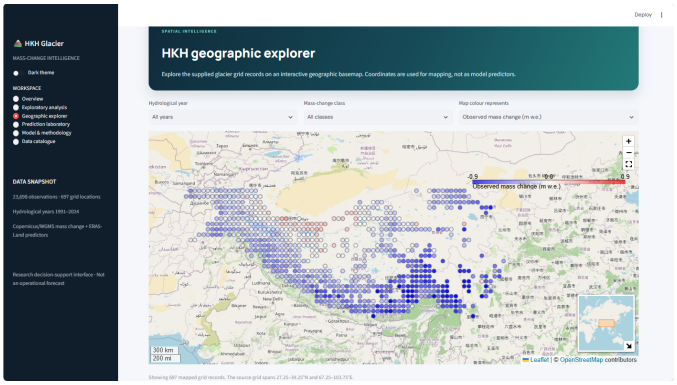


Fig. 32. Streamlit geographic explorer.

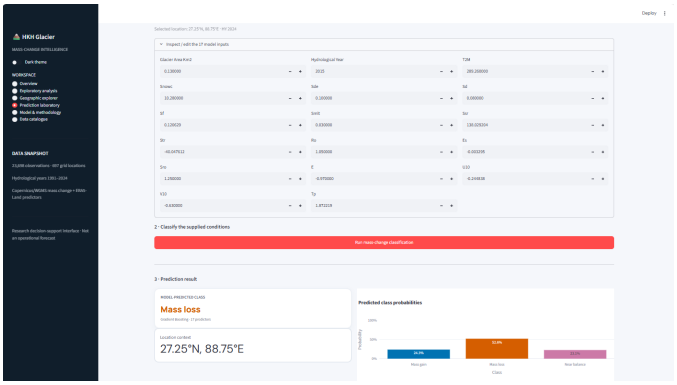


Fig. 33. Streamlit prediction laboratory: input controls and prediction interface.

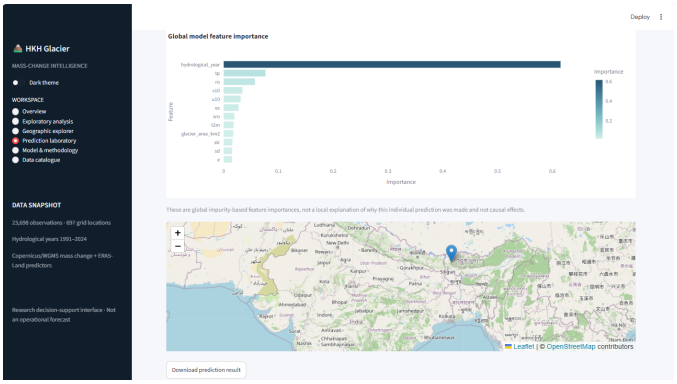


Fig. 34. Streamlit prediction laboratory: model output and probabilities.

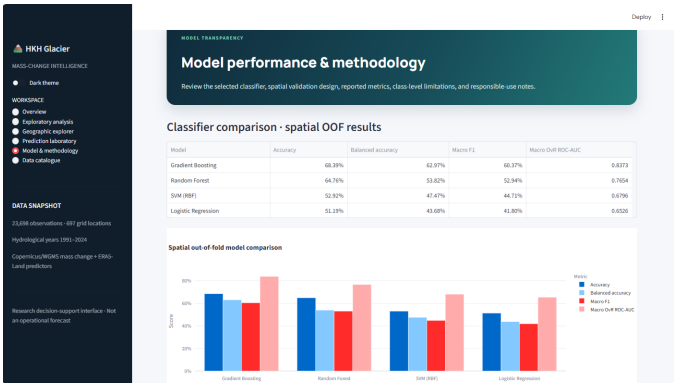


Fig. 35. Streamlit model and methodology page.

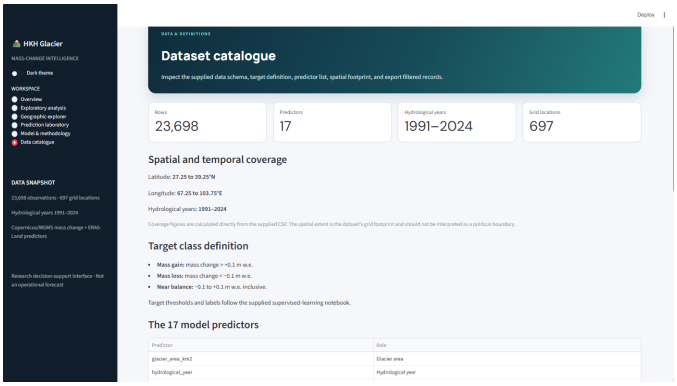


Fig. 36. Streamlit data catalogue page.